

REGULATORY SCORECARD

Bioresorbable Cardiac Stent

Class D · novel · MD-7 (Central) · likely MD-22 / test-licence (MD-13) path

LIKELY CLASSIFICATION

Class D · novel

High complexity

CONFIDENCE

High confidence



LIKELY PATHWAY

MD-7 (Central) · likely MD-22 / test-licence (MD-13) path

Clinical investigation likely required.

TIMELINE + COST BAND

36–60 months

Likely compliance spend: Rs 17–41L

READINESS



Band: Red (significant work ahead)

RISK LEVEL

High

Risk = potential clinical/regulatory exposure today.

WHAT TRIGGERED THIS CLASSIFICATION

Influences clinical management

Novel indication (no Indian predicate)

TOP GAPS TO CLOSE

ISO 10993 biocompatibility evidence not documented

No predicate + no clinical data — MD-26/MD-27 pre-permission and pivotal investigation needed

ISO 13485 QMS not detected

RECOMMENDED NEXT ACTION

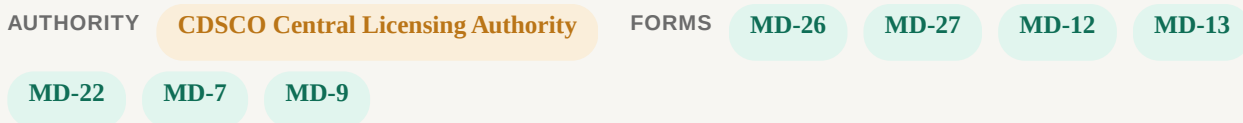
Scope an ISO 10993 testing plan for a long-term blood-contacting implant — minimally -4 (blood interactions), -6 (implantation), -10 (sensitisation), -11 (systemic toxicity), plus degradation/leachables for the bioresorbable polymer. Engage a NABL-accredited testing lab.

SECTION 2

Your Likely Regulatory Pathway

Why this likely class applies to your product — and the form sequence reviewers typically expect.

A bioresorbable cardiac stent sits at the top of the risk ladder under Indian device rules. It is a long-term implant (well beyond 30 days), placed in direct contact with the circulatory system, and intended to support a critical clinical state where failure could be life-threatening. The information it delivers — mechanical scaffolding during vessel healing, then controlled resorption — is highly significant to patient outcomes, and there is no established Indian predicate to lean on. That combination of sustained contact, cardiovascular site, and novel resorbable behaviour typically lands a device in the highest-risk category. The likely route is the manufacturing licence on Form MD-7, with the Central Licensing Authority as the granting body.



LIKELY STEP SEQUENCE

01 MD-26 / MD-27 pre-permission

File the pre-permission application before MD-7 because no Indian predicate is claimed. Adds a separate review cycle.

2–4 MONTHS

02 Test licence (MD-12 / MD-13)

Apply via MD-12 for a test licence (granted as MD-13) so a small number of devices can be made for the clinical investigation.

2–3 MONTHS

TEST LICENCE NOTE

A test licence (granted as MD-13, applied via MD-12) typically precedes the clinical investigation. The investigation itself runs under MD-22 / MD-23 permissions before the device can be cleared for commercial manufacture.

03 MD-22 clinical-investigation permission + EC approval

Design the multi-centre study, get EC clearance, register on CTRI, enrol participants.

9–14 MONTHS

04 MD-7 manufacturing licence (CLA -> MD-9)

File MD-7 with the Central Licensing Authority alongside DMF, PMF, Fifth-Schedule QMS, clinical evidence, biocompatibility (and sterilization validation where applicable). CDSCO MD Officer team conducts a pre-grant site audit within 60 days of application.

6–9 MONTHS

SECTION 3

Readiness Gap Analysis

The gaps a reviewer is likely to flag, ordered by priority. Effort and cost bands are typical Indian medtech ranges — your numbers may be lower with in-house resources or higher with a Big-Four-style consultant.

PRI	GAP	WHY IT MATTERS	SUGGESTED NEXT STEP	EFFORT · COST
P1	ISO 10993 biocompatibility evidence not documented	For a permanent bioresorbable cardiac implant with circulating blood and tissue contact, reviewers will likely expect a full ISO 10993 evaluation matched to long-term implant exposure — typically -1 (overall evaluation), -4 (blood interaction), -6 (implantation effects), -11 (systemic toxicity), -10 (sensitisation/irritation), plus -17/-18 for degradation products and leachables given the resorbable chemistry. Without this evidence packaged into DMF Section 8.11, the file is typically held at first review since biocompatibility is a non-negotiable checklist item for Class D implants.	Scope an ISO 10993 testing plan for a long-term blood-contacting implant — minimally -4 (blood interactions), -6 (implantation), -10 (sensitisation), -11 (systemic toxicity), plus degradation/leachables for the bioresorbable polymer. Engage a NABL-accredited testing lab.	3–6 months · Rs 2–8L
P1	No predicate + no clinical data — MD-26/MD-27 pre-permission and pivotal investigation needed	Because this is a novel Class D implant with no predicate and a resorbable mechanism that behaves differently over time, reviewers will likely require a CDSCO-permitted pivotal clinical investigation in the Indian population — MD-26 permission to conduct the study, EC approvals at each site, CTRI registration, and an MD-27 once data is in hand. This is typically the longest and most expensive item on the critical path, and the submission cannot meaningfully advance until the investigation plan is locked.	Plan a CDSCO pre-submission meeting to align on pre-permission requirements (MD-26/MD-27) and pivotal clinical investigation design (MD-22). Budget for first-in-human and pivotal trials.	9–14 months · Rs 8–18L

P1	ISO 13485 QMS not detected	Reviewers will likely expect a documented ISO 13485 quality system — design controls, CAPA, internal audit records, supplier controls, and process validation — before a Class D manufacturing licence is granted, since implant-grade manufacturing carries the highest QMS scrutiny. Without a certificate or a credible implementation plan with audit evidence, the application is typically held until the quality system can be demonstrated on the shop floor.	Engage an ISO 13485 consultant for a gap assessment and begin building a design-control and risk-management (ISO 14971) framework appropriate for a Class D implantable.	6–9 months · Rs 3–5L
P2	Sterilization validation method not documented	For an implanted cardiac device, reviewers will likely expect a complete sterilization validation aligned to the chosen modality — ISO 11135 (EO), ISO 11137 (radiation), or ISO 17665 (steam) — with bioburden data, SAL 10^{-6} demonstration, dose mapping or biological indicator results, and packaging integrity / shelf-life validation. Resorbable polymers are also sensitive to sterilization conditions, so reviewers typically look for evidence that the chosen method does not compromise the degradation profile; missing data in DMF Section 8.14 typically triggers a query loop.	Pick the method (EtO / gamma / steam / aseptic), engage a validated sterilization partner, and assemble the validation file per the relevant standard (ISO 11135 / 11137 / 17665).	2–4 months · Rs 1.5–3L
P3	Device Master File (Appendix II) not assembled	The Device Master File under Appendix II of the Fourth Schedule is the backbone of an MDR-2017 submission — reviewers cross-check every claim (classification, design, materials, risk file, V&V, biocompatibility, sterilization, stability, clinical evidence, PMS plan, batch CoA) against this single document. Without an assembled DMF, the file typically cannot be accepted for technical review, regardless of how strong the individual studies are.	Assemble the DMF per Appendix II of the Fourth Schedule — descriptive info, classification, product specification, predicate comparison, labelling, design & mfg, essential principles, risk file, V&V, biocomp, sterilization, stability, clinical evidence, PMS, batch CoA.	2–4 months · Rs 1–3L

SECTION 4

Timeline + Cost Estimator

Phased roadmap with Indian-context cost bands. Total range and anchor reflect the readiness card's headline estimate.

TOTAL 36–60 months Class D novel implantable: MD-26/MD-27 pre-permission, pivotal clinical investigation (MD-22), manufacturing licence and post-market surveillance — typically a 3–5 year arc from pre-MVP.

Phase 1 — Foundational compliance Engage an ISO 13485 consultant for a gap assessment, document the Intended Use Statement, and lock the classification rationale.	3–6 MONTHS Rs 3–6L
Phase 2 — Technical documentation Build the Device Master File (Appendix II) — descriptive info, predicate comparison, design & manufacturing, essential principles, ISO 14971 risk file; ISO 10993 biocompatibility evidence at the tier matching patient contact; Sterilization validation per the chosen method (ISO 11135 EO / ISO 11137 radiation / ISO 17665 steam).	4–7 MONTHS Rs 3–9L
Phase 3 — Clinical validation Design and run a multi-centre clinical investigation with EC approval and CTRI registration. Typically the longest item on the critical path.	9–14 MONTHS Rs 8–18L
Phase 4 — Submission preparation Assemble the Plant Master File (Appendix I), finalise the Device Master File, labelling and IFU, predicate or substantial-equivalence narrative, and pre-submission review.	2–3 MONTHS Rs 2–5L
Phase 5 — CDSCO review cycle Submission filing (MD-3 to SLA or MD-7 to CLA depending on class), queries-and-responses cycle, pre-grant site audit (60 days for Class C/D, 90 days for Class B), licence grant.	4–8 MONTHS Rs 1–3L

LIKELY BOTTLENECKS

- Clinical investigation timeline (EC + CTRI + enrolment + follow-up) dominates the schedule.
- Top high-severity gap: ISO 10993 biocompatibility evidence not documented.
- Novel device path adds MD-26 / MD-27 pre-permission ahead of MD-7.

SECTION 5

Reviewer Insights

What CDSCO reviewers will likely look for in a submission of this shape. Generic regulatory checklists rarely capture reviewer psychology — this is yours.

Strength of clinical validation evidence

For a novel Class D bioresorbable cardiac stent, reviewers will likely expect a prospective Indian clinical investigation under NDCT Rules 2019, with Ethics Committee clearance, CTRI registration, and a pre-specified SAP. Endpoints such as TLF, scaffold thrombosis, and late lumen loss typically receive line-by-line scrutiny, alongside imaging core-lab adjudication, resorption-timeline data, and adequate follow-up duration to characterise the bioresorption profile.

Intended Use consistency across documents

Reviewers typically cross-check the Intended Use Statement against the IFU, labelling, brochures, and website copy. For a bioresorbable cardiac stent, common drift areas include vessel-size range, lesion type, patient eligibility, and resorption-claim wording. Marketing language hinting at benefits beyond the validated indication — for example off-label lesion subsets or comparative claims versus DES — may trigger early clarification queries.

Biocompatibility tier matching patient contact

As a long-term implant in circulating blood, the stent likely sits in the most stringent ISO 10993-1 tier. Reviewers may expect 10993-4 (haemocompatibility), 10993-6 (implantation, with histopathology across resorption timepoints), 10993-10 (sensitisation, irritation), 10993-11 (systemic and sub-chronic toxicity), and a detailed 10993-17/-18 leachables/extractables profile covering polymer degradation products and any drug-coating residues.

Indian-population validity

Given novelty and likely reliance on overseas trial data, reviewers typically ask how performance translates to Indian patients — vessel-size distribution, diabetes and CAD pattern differences, younger STEMI demographics, and dual-antiplatelet adherence context. Subgroup analyses by age, sex, diabetic status, and lesion complexity, along with site-level variability across Indian cath labs and imaging protocols, may receive close attention.

Cybersecurity & data handling

Cybersecurity scrutiny is typically limited for a passive bioresorbable stent without connected components. If trial data, imaging, or registry PHI are handled digitally, reviewers may ask for a basic data-protection narrative — encryption in transit and at rest, role-based access, DPDP Act 2023 alignment, and CERT-In safe-to-host evidence for any cloud-hosted CTMS or core-lab imaging platform supporting the clinical investigation.

SECTION 6

Smart Examples

Annotated good-vs-bad wording. These are snippets, not full forms — copy them as models for your own Intended Use, claim, and risk-justification language.

INTENDED USE Concrete user + setting + function

GOOD

Intended for use by interventional cardiologists in cath-lab settings for percutaneous implantation in adult patients with de novo native coronary artery lesions, as an adjunct to optimal medical therapy and standard angiographic guidance. The bioresorbable scaffold provides temporary mechanical support and is not a substitute for clinician judgement on lesion selection or sizing.

BAD

A next-generation dissolving stent that helps patients recover from heart blockages naturally without permanent implants.

WHY THIS IS SAFER

The first wording names the user, the setting, the patient group, the lesion type, and the adjunctive role — all verifiable against the IFU and clinical investigation plan. The second is consumer marketing copy and gives reviewers nothing concrete to cross-check.

CLAIM WORDING Indication scope vs. label creep

GOOD

Indicated for improving coronary luminal diameter in adult patients (age 18+) with a single de novo native coronary artery lesion, reference vessel diameter 2.5-3.75 mm and lesion length up to 24 mm, when treated by a qualified interventional cardiologist. The device is not validated for in-stent restenosis, bifurcation lesions, left main disease, bypass grafts, or paediatric use.

BAD

Suitable for treatment of coronary artery disease and related ischaemic conditions in adults.

WHY THIS IS SAFER

The first wording names the lesion type, vessel dimensions, operator, and explicit exclusions — matching what a clinical investigation can realistically validate. The second is broad enough to invite label-creep questions on bifurcations, left main, and restenosis that the evidence likely will not cover.

RISK JUSTIFICATION Residual risk acceptance

GOOD

After applying the documented design controls (strut thickness specification, resorption profile testing per ISO/TS 17137, and operator training requirements), the residual risk of late scaffold thrombosis is judged acceptable in light of the device's clinical benefit, supported by 12-month MACE data from the pivotal clinical investigation. Residual risk is monitored under the PMCF plan, with six-monthly review of thrombosis and target-lesion failure signals.

BAD

All residual risks from scaffold resorption and thrombosis have been accepted as low and require no further action.

WHY THIS IS SAFER

The first wording cites specific controls, a named standard, clinical evidence, and a PMCF monitoring cadence — the structure ISO 14971 expects for a Class D implant. The second is a bare assertion and invites reviewers to ask for the underlying acceptance criteria and surveillance plan.